II. Configuring VM in VMWare Workstation

(DELL PC | Windows 10)

**For this setup, I am using Windows 10 and VMWare Workstation 17 Pro 17.6.4 build-24832109 **

OS I used was: ubuntu-24.04.3-desktop-amd64.iso

Section 1: Set Up Virtual Network Adapters

this will disconnect your computer from the subnets

a. *Edit>Virtual Network Editor…*
b. *Change Settings*
    i. run this menu as administrator
c. Configure existing VMNets
    i. VMNet0
        - Change to Host-Only
    ii. VMNet1
        - Leave as is
    iii. VMnet8
        - Delete, makes organization easier
d. Create new VMnet's
    i. For 10.10.10.1 Click *Add Network*
        - *Select VMnet2*
        - In VMnet Information
            - Set to *Bridged*
            - Find the adapter you want to connect to thats connected to 10.10.10.1 Network
    ii. For 10.10.20.1 Click *Add Network*
        - *Select VMnet3*
        - In VMnet Information
            - Set to *Bridged*
            - Find the adapter you want to connect to thats connected to 10.10.20.1 Network
    iii. For 10.10.30.1 Click *Add Network*
        - *Select VMnet4*
        - In VMnet Information
            - Set to *Bridged*
            - Find the adapter you want to connect to that's connected to 10.10.30.1 Network
    iv. For computer LAN internet connection Click *Add Network*
        - *Select VMnet8*
        - In VMnet Information
            - Set to *NAT*
            - Check the *Connect a host virtual adapter to this network*
            - Check the *Use local DHCP service*

2. **Spin-Up VM**
    a. Right Click Over “Library”>New Virtual Machine…
    b. In *New Virtual Machine Wizard*
        i. Typical (Recommended)
        ii. Find boot file
        iii. Make Full Name, User name, Password
        iv. Make a VM Name and choose a file location for VM
        v. 20GB Disk
    c. In *Customize Harware…* menu
        i. Memory: 8GB
        ii. Processors: 2
        iii. New CD/DVD (SATA): leave as is
        iv. USB Controller: Present
        v. Sound Card: Auto Detect
        vi. Display: Auto Detect
    skip network adapters quick
    d. Network Adapters
    For New Network Adapters Click *Add…>Network Adapter*
        i. Network Adapter 2
            - Set adapter to Custom: Specific Virtual Network
                - Select VMnet for NAT
            - Click *Advanced…*
                - Generate a MAC Address and set the last prefix to “:01”
        ii. Network Adapter 2
            - Set adapter to Custom: Specific Virtual Network
                - Select VMnet for 10.10.10.1 subnet
            - Click *Advanced…*
                - Generate a MAC Address and set the last prefix to “:02”
        iii. Network Adapter 3
            - Set adapter to Custom: Specific Virtual Network
                - Select VMnet for 10.10.20.1 subnet
            - Click *Advanced…*
                - Generate a MAC Address and set the last prefix to “:03”
        iv. Network Adapter 4
            - Set adapter to Custom: Specific Virtual Network
                - Select VMnet for 10.10.30.1 subnet
            - Click *Advanced…*
                - Generate a MAC Address and set the last prefix to “:04”
    e. Click *Finish* and complete boot setup for Linux Ubuntu

3. Configure Network on VM

a. Open terminal on Ubuntu Desktop

b. Check ip Links

```
ip link show
```

This will show you if all your links are present

Keep Track of ens_ numbers and corresponding MAC addresses

c. Configure Static IPs

```
sudo nano /etc/netplan/01-router.yaml
```

Paste/Type this into the the *01-router.yaml* document:

Beware of ens numbers as other VM's might assign different numbers

```
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      addresses: [10.10.10.1/24]
    ens34:
      addresses: [10.20.20.1/24]
    ens35:
      addresses: [10.30.30.1/24]
    ens36:
      dhcp4: true
```

d. Apply changes

```
sudo netplan apply
```

e. Check work

i. Ping 10.10.10.1

ii. Ping www.google.com

```
ip -br addr show
```

4. Enable SSH (optional)

This is so you can SSH from the Windows Desktop into your VM and then to the Pi's

a. Install the SSH Server Program

```
sudo apt update
sudo apt install openssh-server -y
```

b. Enable SSH Server

```
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
```

5. Turn VM into a router

a. Confirm you can ping Pi's
b. SSH into Pi's and see what gateways/devices you can ping
c. Paste these commands in to enable port forwarding

For Internet:

```
sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
```

MEDIA-to-Internet:

```
sudo iptables -A FORWARD -i ens34 -o ens33 -j ACCEPT
sudo iptables -A FORWARD -i ens33 -o ens34 -m state --state
RELATED,ESTABLISHED -j ACCEPT
```

CONTROL-to-Internet:

```
sudo iptables -A FORWARD -i ens35 -o ens33 -j ACCEPT
sudo iptables -A FORWARD -i ens33 -o ens35 -m state --state
RELATED,ESTABLISHED -j ACCEPT
```

MANAGEMENT-to-Internet:

```
sudo iptables -A FORWARD -i ens36 -o ens33 -j ACCEPT
sudo iptables -A FORWARD -i ens33 -o ens36 -m state --state
RELATED,ESTABLISHED -j ACCEPT
```

For VLAN-to-VLAN

```
sudo iptables -A FORWARD -i ens34 -o ens35 -j ACCEPT
sudo iptables -A FORWARD -i ens35 -o ens34 -j ACCEPT
sudo iptables -A FORWARD -i ens35 -o ens36 -j ACCEPT
sudo iptables -A FORWARD -i ens36 -o ens35 -j ACCEPT
```

d. Confirm the Rules are in Place

```
sudo iptables -L -v
sudo iptables -t nat -L -v
```

e. Make them persistent

```
sudo apt install iptables-persistent -y
sudo iptables-save | sudo tee /etc/iptables/rules.v4
```

f. Test routes from VM and from Pi's